



Quantum causal models: the merits of the spirit of Reichenbach's principle for understanding quantum causal structure

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Abstract

Through the introduction of his ‘common cause principle’ [*The Direction of Time*, 1956], Hans Reichenbach was the first to formulate a precise link relating causal claims to statements of probability. Despite some criticism, the principle has been hugely influential and successful—a pillar of scientific practice, as well as guiding our reasoning in everyday life. However, Bell’s theorem, taken in conjunction with quantum theory, challenges this principle in a fundamental sense at the microscopic level. For the same reason, the celebrated causal model framework pioneered by Pearl, as well as by Spirtes, Glymour and Scheines, defies satisfactory causal explanations of quantum correlations—the role of the ‘causal Markov condition’ in this framework amounts to a generalisation of Reichenbach’s principle. Much effort has been devoted to the development of quantum causal models to overcome this challenge and to study whether a principled way of causal reasoning, albeit adjusted in its terms, can be maintained when it comes to quantum physics. A clarification of what quantum causal relations are supposed to be is also a much needed step in light of the hotly debated topic of causality in foundations of quantum theory, e.g. in the study of ‘indefinite causal structures’. This paper reports and discusses a framework of quantum causal models and argues that it is promising and interesting for two main reasons. First, it can be seen as a prime example of a recent research programme that took direct and fruitful inspiration from ideas first formulated by Reichenbach—it began by asking, and giving an answer to, how the common cause principle could be generalised appropriately to quantum theoretical terms. Second, the framework did not only facilitate generalisations of the core theorems of the classical framework of causal models, but, above all, it has a theorem at its center that justifies the ‘quantum causal Markov condi-

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tion’ on the basis of what arguably is a natural definition of causal relations in quantum theory. This provides a conceptually clear grounding of the framework, as well as a proposal for the needed clarification of quantum causal relations in the foundations of quantum theory more generally.

Keywords Reichenbach’s common cause principle · Quantum theory · Causal reasoning · Bell’s theorem · Causal models · Quantum causal models

1 Introduction

If causal reasoning is to be part of the scientific methodology, a precise link between claims in causal terms and claims in empirical terms, typically statements of a statistical nature, is required. Hans Reichenbach was one of the first to study such a link and to formulate a corresponding bridge principle—his *common cause principle* (Reichenbach, 1956)—albeit as part of a particular, wider philosophical programme in his *The Direction of Time* (Reichenbach, 1956) (see, e.g., Glymour & Eberhardt, 2016; Hitchcock & Rédei, 2020 for a discussion). Using the terms of contemporary discussions, the principle’s essence may be paraphrased as: suppose two variables Y and Z are correlated, i.e. $P(Y, Z) \neq P(Y)P(Z)$, then there has to exist a causal explanation of this fact and if none is a cause of the other, then there furthermore has to exist a common cause X such that, when conditioning on X , the variables Y and Z are rendered independent, i.e. $P(Y, Z|X) = P(Y|X)P(Z|X)$.

This principle may be seen to underlie much causal reasoning in everyday life and is also a pillar of scientific practice. Indeed, the principle finds its generalisation, both in spirit and in a technical sense, in the framework of *causal models*, pioneered in particular by Pearl (2009) and by Spirtes et al. (2000). There, a general causal structure—the set of causal relations among a set of variables—as represented by a directed acyclic graph (DAG), is asserted to leave a signature in a probability distribution over the same variables in form of the *causal Markov condition*. This way of making precise how, in the general case, causal claims relate to probabilistic statements (ignoring here the problem of going from finite data to probabilities) facilitated ‘causal discovery algorithms’, which spell out, in a principled way, under what assumptions the causal structure can be inferred from observational data. Causal models also allow one to make predictions, whenever that is possible at all, about what would happen if one *intervened* and e.g. forced a variable to take a certain value—a question about which a probability distribution alone tells us nothing. A particular approach to causal models, called *structural equational models*,¹ further provides a logic for evaluating counterfactual statements. For all these reasons, the framework has attracted much attention from applied fields, such as medicine and economics, as well as from philosophy (Hitchcock, 2020; Pearl & Mackenzie, 2018).

While various alleged counterexamples and other challenges to Reichenbach’s common cause principle have been put forward and debated in the literature (Hitchcock

¹ Sometimes also called ‘structural causal models’ (Bareinboim et al., 2020) or ‘functional models’ (Pearl, 2009).

& Rédei, 2020), its empirical success, in particular in its generalised form relevant to causal models, arguably still stands. However, one particular challenge to the principle and to the range of validity of the causal model framework weighs heavily, as it concerns empirical facts that are in accordance with quantum theory—one of the best fundamental theories of physics we have. This is the fact that, due to Bell's theorem (Bell, 2004), correlations that violate a Bell inequality, in particular EPR correlations, defy a common cause explanation in terms of only free choices of measurement at space-like separated regions and a variable in the common past, as the principle would demand. Quantum theory, however, predicts precisely such correlations and these have also been verified in numerous experiments. Hence, the principle seems falsified. There are subtleties in this argument, which must not be overlooked (see, e.g., Hitchcock & Rédei, 2020; Myrvold et al., 2020 for overviews), but it is safe to note that any strategy of retaining a causal explanation of Bell correlations in terms of causal models with classical variables as relata is doomed to be fine-tuned (Wood & Spekkens, 2015; Cavalcanti, 2018)—the framework of *classical* causal models does not provide the grounds to give satisfactory causal explanations of quantum correlations.

There has been much work on 'causality' in quantum foundations and quantum information theory that is in some sense or other explicitly concerned with causal structure (more on this later). In particular, much effort has been put into developing a quantum generalisation of causal models that would make it possible to give causal accounts of quantum phenomena (see, e.g., Costa & Shrapnel, 2016; Fitzsimons, 2015; Fritz, 2016; Henson et al., 2014; Laskey, 2007; Leifer, 2006; Leifer & Poulin, 2008; Leifer & Spekkens, 2013; Pienaar, 2019, 2020; Pienaar & Brukner, 2015; Pienaar, 2019; Ried et al., 2015; Tucci, 1995, 2007). This suggests that researchers see causal reasoning as, at least potentially, meaningfully applicable in quantum contexts. However, what arguably has been missing from these accounts is a conceptual foundation for quantum causal reasoning. Proposed answers about what the relata are, what the causal connection is, and how to give causal explanations in the most general, yet principled way have remained inconclusive.

In this paper, we report on a promising recent research programme that addresses these questions. While the full programme and its results can be read elsewhere (Allen et al., 2017; Barrett et al., 2019, 2021; Lorenz & Barrett, 2021), the aim of this paper—a development of a talk given at the conference *All Things Reichenbach* in Munich, 2019—is to raise for cross-disciplinary consideration this interesting example of Reichenbach's legacy in contemporary science. The author (a contributor to the research programme that will here be discussed) has no formal philosophical training, but wishes to make the case for fruitful further study from both technical and conceptual perspectives, inviting critical analysis and proposing a topic of further consideration in philosophy of physics and causation.

The research programme in question produced a framework of quantum causal models that, we argue, is a good candidate for formalising causal reasoning about quantum systems (although so far developed only for finite-dimensional systems in quantum theory). It has led to some noteworthy technical results and further developments, but—most importantly—involves a proposal for how to define *direct-cause relations* in quantum theory and grounds the quantum causal model definition in this. These developments started with the work by Allen et al. (2017), which analyses

Reichenbach's common cause principle and how its spirit can be translated to a common cause principle in quantum theoretical terms—concerning both, explanans and explanandum—which then led the way to a generalisation to arbitrary causal structures.

This paper provides a brief summary and discussion of this framework of quantum causal models. It presents the definitions of the main concepts—trying to make them accessible and intuitive without going into the weeds of the quantum formalism—and analyses and comments on the developments in what is hoped to be a fruitful way. As it aims to tease out fruitful connections and relevance, rather than set out fully the details of the framework, which has already been done in Barrett et al. (2019), it is inevitably somewhat limited in precision and depth, but it aims to provide enough substance to reveal the philosophically interesting aspects of the work.

With this in mind, the paper has the following structure. Section 2 elaborates on the motivation by recapitulating and fleshing out some of the points mentioned in this introduction. Section 3 does a cursory ride through definitions and simply states the classical and quantum versions of causal models and causal principles, in close analogy to each other, to set out the terms and to substantiate the claimed link to Reichenbach's principle. Section 4 sketches the conceptual heart of the research programme—the definition of causal relations and the justification of the quantum causal principles. Sections 5 and 6 clarify some naturally arising questions and comment with some reflections. Section 7 concludes.

2 Motivation and background

The problem of Bell correlations

Quantum systems can give rise to correlations that appear to evade a causal account—at first sight anyway. In order to sketch the essence of the standard argument as to why this is the case, suppose a 'Bell state', i.e. a maximally entangled bipartite state, say, a pair of electrons with their spins in state $(1/\sqrt{2})(|00\rangle + |11\rangle)$, is prepared and the two subsystems are then sent off to space-like separated regions B and C , where respective agents perform measurements with outcomes O_B and O_C . For appropriate choices M_B and M_C of their respective measurements (say spin S_X or S_Z), and upon comparing their measurement statistics, they find that $P(O_B, O_C|M_B, M_C) \neq P(O_B|M_B) P(O_C|M_C)$, where it was used that O_B must not depend on M_C and similarly, O_C not on M_B , for it to be in keeping with relativity. By the same token the space-like separated O_B and O_C must not be causes of each other.

Reichenbach's common cause principle then demands there exist a common cause variable λ in the region common to the two past light cones of B and C , such that $P(O_B, O_C|M_B, M_C, \lambda) = P(O_B|M_B, \lambda) P(O_C|M_C, \lambda)$ holds. However, Bell's theorem (Bell, 2004) basically states that from the latter expression an inequality can be derived, which in fact is violated by the predicted—and empirically observed—conditional probabilities $P(O_B, O_C|M_B, M_C)$. First, this highlights the role of Reichenbach's principle in Bell's condition of 'local causality' that Bell argued

to codify what a satisfactory scientific explanation of correlations is (see, e.g., Myrvold et al., 2020; Wiseman & Cavalcanti, 2017 for discussions). Second, given the Bell-inequality violations seen in numerous experiments, Reichenbach's principle seems to fail as—and this is the standard conclusion from the theorem—there cannot, in principle, exist a variable in the common past of B and C that would warrant an appropriate common cause explanation in keeping with the principle. Because the principle essentially instantiates the special case of a common cause explanation within the causal model framework, the latter framework as a whole must thereby also be deemed to fail at giving an account of quantum correlations.

Of course, Bell inequality violations only allow one to infer the failure of the conjunction of all assumptions that went in, and there are various strategies to retain an explanation of Bell correlations in terms of 'classical' variables, such as invoking superdeterminism, retrocausal influences or superluminal causal influences. These can even be cast as causal model explanations, however, with the drawback of requiring some form of fine-tuning (Cavalcanti, 2018; Wood & Spekkens, 2015). The main point hence stands: the framework of classical causal models does not provide the grounds to give satisfactory causal explanations of quantum correlations.

Possible lessons from the issue and directions to take

One response is to deem causal reasoning to pertain only to the macroscopic world, well described by classical probability theory, concluding that there just is no place for it at the level of quantum physics. While one may take such an attitude, it seems at the very least hasty to conclude it is the only route for a concept so deeply bound up with common intuitions about the world and good explanations. What if one *can* arrive at an adjusted kind of causal reasoning suitable for quantum physics, by dropping some of the aspects and terms usually associated with it, while keeping other, hopefully essential aspects?

Closely related to these questions lies a mature research programme with many strong results due to Rédei, Hofer-Szabó and their collaborators (see, e.g., Gyenis & Rédei, 2011; Gyenis & Rédei, 2016; Hofer-Szabó & Rédei, 2006; Hofer-Szabó & Vecsernyés, 2012, 2013a, 2013b, 2018; Kitajima & Rédei, 2015; Rédei, 1997). One of the questions it has been investigating is which (generalised) probability spaces have the property to be *common cause complete*, meaning that every correlation has a Reichenbachian common cause that could explain it—at least *in principle*, that is, ignoring any further constraints from a particular theory or from what sort of thing we would be happy to accept as a cause. More generally, a space may be *common cause completable* if embeddable into a larger one that is complete. The idea is, if a space is proven to *not* be completable, the principle is falsified, at least on the basis of the generalised notion of a Reichenbachian common cause that was considered. These efforts have in particular revealed many subtleties in the above crude sketch of how Bell inequality violations are taken to imply a failure of Reichenbach's common cause principle (see Hitchcock & Rédei, 2020 for a good summary of the relevant results; also see Cavalcanti & Lal, 2014). One might then ask whether the kind of quantum probability space associated with finite-dimensional non-relativistic quantum theory is common cause completable—contemplating a quantum Reichenbach principle. First, the results in Hofer-Szabó and Vecsernyés (2013b) answer this in the negative, or at

least severely narrow it down (depending on the assumptions one is happy to commit to). Second, the underlying notion of a common cause in these works has been critically analysed by Cavalcanti and Lal (2014) concerning the explanatory work it can do at all.

From the perspective of quantum theory, in the Bell scenario described above there of course is an obvious candidate of a common cause many would like to point to—the preparation of a Bell state in the common past; an interaction between the two subsystems has to take place to bring about the sorts of correlations that can later violate a Bell inequality. The question then is whether there can be a quantum generalisation of causal reasoning that marks a more radical departure from Reichenbach's principle than looking for Reichenbachian common causes in a generalised probability space—a generalisation that looks for kinds of explanandum and explanans that are appropriately lifted to quantum theoretical terms, while still preserving some essence of the common cause principle (for what is the point otherwise?). Wood and Spekkens (2015), as well as Cavalcanti and Lal (2014) argued for such hope. A *quantum causal model* framework would be one that answers that question affirmatively and succeeds in making the mentioned intuition on the Bell scenario precise in a general and principled way.

Further motivation from physics

Increasingly over the past two decades, research on 'causality' has been attracting much interest and attention in quantum foundations and quantum information theory. Among physicists the term 'causality' is an overloaded term, which is often used to refer to constraints implied by the structure of spacetime—'causality' in this sense is about constraints on what can possibly be causally related to what to be in keeping with relativity. These constraints as such are ignorant of any theory of causation; they do not tell us how causal relations are defined, in particular not concerning quantum theory. (See, e.g., Vilasini & Renner, 2022 for a discussion of the different notions of causality in physics.)

The research activity on causality in physics is broad and spans both, relativistic causality just mentioned, as well as questions more concerned with what causation is and how to give causal explanations. Here causation tends to be understood more or less explicitly as the possibility of 'signalling'. To mention but a tiny fraction of relevant aspects and in each case just giving a small selection of respective works, these include the study of the no-signalling constraints from relativistic causality on generic quantum evolution (Arrighi et al., 2011; Barrett et al., 2005; Beckman et al., 2001; Schumacher & Westmoreland, 2005), the derivation of Bell-type inequalities for generalised causal scenarios (Chaves et al., 2015; Fritz, 2016; Henson et al., 2014; Pienaar, 2017; Wood & Spekkens, 2015) and 'causality' as an axiom or property of a theory, which could be paraphrased as 'no signalling back from the future' and which plays a prominent role in, e.g., reconstructions of quantum theory and elucidating in a general way quantum theory's relation to relativistic causality (Chiribella et al., 2010, 2011; Coecke 2014; Coecke & Kissinger, 2017; Coecke & Lal, 2013; D'Ariano et al., 2017; Kissinger et al., 2017; Selby et al. 2021).

The idea of 'quantum systems as causes' has been an influential intuition in many of these works. Further developing this intuition, there have in particular been all the

works on a quantum generalisation of classical causal models, already mentioned in the introduction (Costa & Shrapnel, 2016; Fitzsimons et al., 2015; Fritz, 2016; Henson et al., 2014; Laskey, 2007; Leifer, 2006; Leifer & Poulin, 2008; Leifer & Spekkens, 2013; Pienaar, 2019, 2020; Pienaar & Brukner, 2015; Ried et al., 2015; Tucci, 1995, 2007). These works, while containing many important ideas and results as stepping stones, were somewhat piecemeal, each either lacking generality, or a conceptually clear picture on causal relations and why the respective causal model definition ought to be as proposed, or else both. For a more detailed analysis of these earlier works and their comparison with the framework that this paper focuses on, in particular in the case of the closely related work by Costa and Shrapnel (2016), the reader is referred to Allen et al. (2017); Lorenz (2020) and Barrett et al. (2019).

Finally, it should be noted that working towards a clarification of quantum causal relations and principled causal explanations seems especially pertinent in light of a particularly active field of research, namely, that of studying the possibility, realisability and signature of ‘(quantum) indefinite causal order’ with motivation coming from quantum gravity and more abstractly ‘quantum superpositions of causal order’ (see, e.g., Araújo et al., 2014, 2017; Baumeler et al., 2014; Bisio & Perinotti, 2019; Brukner, 2014; Chiribella et al., 2013; Guérin et al., 2016; Hardy, 2005; Hardy, 2007; Oreshkov et al., 2012; Oreshkov, 2019; Paunković & Vojinović, 2020; Uijlen & Kissinger, 2019; Wechs et al., 2019 for a small selection). One had better first get a firm grasp of causal relations, where there is a supposed ‘definite causal order’, before moving to an area that is challenging due to further puzzling aspects.

3 From causal models to causal principles and back

The following will present the definitions from the quantum causal model (QCM) framework as proposed and studied in Allen et al. (2017) and Barrett et al. (2019) together with the analogous definitions from the classical causal model (CCM) framework by following the journey as in Fig. 1 via Reichenbach’s common cause principle. This is the historical journey it took—obviously not in the sense that CCMs would precede Reichenbach, but in the sense that this proposal for the quantum generalisation of causal models was enabled by first taking a step back and focusing on the relation between causal models and the basic common cause scenario.

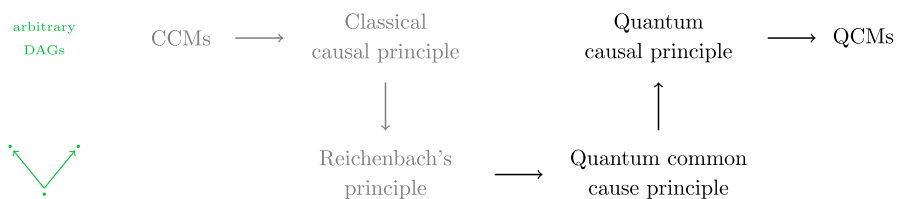


Fig. 1 The journey schematically

3.1 The classical definitions: a blueprint

Let us recall standard definitions from the framework of causal models, though presented in a way so as to carve out a template for our purpose and to ease later reference. (For a detailed introduction and more in-depth explanation the reader is referred to Pearl, 2009; Spirtes et al., 2000, as well as Bareinboim et al., 2020; Eberhardt, 2009; Hitchcock, 2020.) Any random variable X , also referred to simply as a (classical) variable, is throughout assumed to be valued in a set of finite cardinality and $P(X_1, \dots, X_n)$ denotes a probability distribution² over n variables. Causal structure is represented by DAGs, where an arrow from X to Y stands for a *direct-cause relation* from X to Y , while a directed path from X to Y of length more than one is interpreted as that X is an indirect cause of Y .³ A key concept of the framework then is the *Markov condition*, which in this context is often also referred to as the *causal Markov condition*, in order to emphasise that the DAG is given a causal reading.

Definition 3.1 (Classical Markov condition): A probability distribution $P(X_1, \dots, X_n)$ is Markov for a DAG G with vertices $X_1, \dots, X_n$ iff $P(X_1, \dots, X_n) = \prod_i P(X_i | Pa(X_i))$ with $Pa(X_i)$ denoting the parental sets in G .

There are several conditions that are equivalent to the one in Def. 3.1; a particularly intuitive one being that for every variable X_i its parents $Pa(X_i)$ ‘screen it off’ from its other non-descendants, i.e. X_i is conditionally independent from its other non-descendants given its parents.

The basic idea now is that, given variables $X_1, \dots, X_n$ and a probability distribution $P(X_1, \dots, X_n)$ with all the correlations between subsets of the variables it entails, a causal explanation according to the causal model framework consists in stating the direct-cause relations as summarised in a DAG G , demanding that $P(X_1, \dots, X_n)$ be Markov for that G . Hence, the definition:

Definition 3.2 (Classical causal model⁴ (Pearl, 2009; Spirtes et al., 2000)): A (classical) causal model (CCM) is given by:

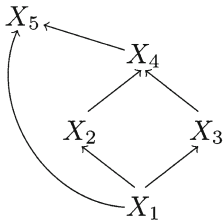
- (1) a causal structure represented by a DAG G with vertices corresponding to random variables $X_1, \dots, X_n$
- (2) a probability distribution $P(X_1, \dots, X_n)$ that is Markov for G .

Importantly, note that a causal model can be seen as in keeping with the following principle, which for the purpose of this paper’s exposition we call the general *classical causal principle*:

² Here assumed to feature only non-vanishing probabilities. Distributions that do not have full support actually play an important role in the framework, conceptually speaking, but this does not change anything this paper focuses on.

³ At least typically, since it is in principle possible that, loosely speaking, the influences along two distinct directed paths from X to Y precisely cancel each other. The assessment of whether X is an indirect cause of Y thus is subtle, see Pearl (2009) and Beebe et al. (2009). Also note that the distinction between a direct and an indirect causal relation is a concept that is dependent on the chosen set of variables.

⁴ In the literature causal models are sometimes also referred to as ‘causal Bayesian networks’ or ‘graphical causal models’. The difference to an arbitrary Bayesian network, i.e. the difference from asserting the DAG to represent causal knowledge, can be made precise, e.g., in terms of predictions for what would happen if one intervened on some of the involved variables.



$$\begin{aligned}
 P(X_1, X_2, X_3, X_4, X_5) = & \\
 P(X_5|X_1, X_4) & P(X_4|X_2, X_3) \\
 P(X_2|X_1) & P(X_3|X_1) \\
 P(X_1) &
 \end{aligned}$$

Fig. 2 Generic example of a classical causal model

- CP* Given a set of variables $X_1, \dots, X_n$ and a probability distribution $P(X_1, \dots, X_n)$,
- (0) there is a causal structure of the variables $X_1, \dots, X_n$, possibly involving other (latent) variables as further common causes.
 - (1) if the set $\{X_1, \dots, X_n\}$ is causally sufficient and the causal structure is given by the DAG G with vertices $X_1, \dots, X_n$, then $P(X_1, \dots, X_n)$ is Markov for G .

The principle first simply asserts that every observed probability distribution will have a causal explanation through the causal relations that actually obtain amongst the relevant variables—a stipulation that is usually left implicit, but here made explicit for our purposes and therefore labelled the 0th part of the principle. The property of *causal sufficiency* of a set S of variables that appears in part (1), means that any further variable that is a common cause to some of those in S is *not a direct* cause to more than one variable in S —since if it were, it had better be included in the causal structure for it to constitute a causal explanation of $P(X_1, \dots, X_n)$.

One use case of this part (1) naturally is in its contrapositive form for causal inference: if a given distribution $P(X_1, \dots, X_n)$ is *not* Markov for some DAG G then G can be ruled out as a plausible causal explanation. It should of course be noted that the causal discovery algorithms in the extensive literature on causal models (see Pearl, 2009; Spirtes et al., 2000 and references therein) rely on many further substantial ingredients than just *CP*, for the Markov condition is a necessary but in general insufficient condition for a DAG to be regarded the causal explanation for a $P(X_1, \dots, X_n)$.⁵ These additional ingredients include both technical results, as well as Occam’s razor type of criteria like ‘faithfulness’,⁶ in order to reduce the typically huge space of DAGs for which a given distribution is Markov and which cannot be ruled out on logical grounds. Allowing for potential latent, unobserved common causes not included in $X_1, \dots, X_n$ that the observational data covers, requires yet more elaborate techniques. However, these details, or for that matter other aspects and tools of the framework such as those that help answer questions of an interventional or counterfactual nature, are not the point of this exposition.

⁵ See, for instance, Hitchcock (2020, 2018) and Eberhardt (2009) for a discussion of the status of the (causal) Markov condition.

⁶ A distribution P is said to be *faithful* to a DAG G iff P satisfies only those conditional independence relations that are implied by G through Markovianity, i.e. those that hold by necessity if asserting G to be the causal structure, while there must not be any further, extraneous conditional independence relations that are not entailed by G . In more technical terms, faithfulness amounts to a 1-to-1-correspondence between the conditional independence relations in P and the d-separation relations in G (see, e.g., Pearl, 2009).

Instead, simply note that *CP* contains *Reichenbach's common cause principle* as a special case, when put as follows (inspired by Allen et al., 2017; Cavalcanti & Lal, 2014):

RCCP Given two random variables Y and Z that are correlated, i.e. $P(Y, Z) \neq P(Y)P(Z)$ ⁷,

- (0) there is a causal explanation of this fact, possibly involving a further variable X , through either (a) $Y \rightarrow Z$, (b) $Y \leftarrow Z$, or (c) $Y \leftarrow X \rightarrow Z$, or else (c) in conjunction with (a), or (c) in conjunction with (b).
- (1) if X is a complete common cause, i.e. no further direct common cause is missing and (a) and (b) are false, then it holds that $P(Y, Z|X) = P(Y|X)P(Z|X)$ ⁸.

The suggested reading here is that a successful causal model framework relies on, and thereby is an expression of, having gotten in particular this link between causal and probabilistic claims in the most basic of cases, the ‘common cause scenario’, right.

3.2 The quantum definitions

The general idea pursued in Allen et al. (2017) is to take the above as a ‘blueprint’ and go the other way, that is, ask what a quantum generalisation of the common cause principle should be that preserves the spirit of Reichenbach’s principle, and then generalise that to arbitrary DAGs, leading the way to quantum causal models.

3.2.1 The quantum common cause principle

What amounts to the ‘spirit’ of *RCCP* can of course be taken to be different things. Here, this spirit is taken to be a weak reading of it—a common cause principle that is obtained by answering the following set of questions:

1. What sort of things are the causal relata?
2. If asserting A is a complete common cause of B and C , what sort of mathematical object, associated with an empirical description of the matter of affairs, is constrained by that causal assertion?
3. What exactly is that constraint?

The below sketches the answers that were first proposed by Allen et al. (2017)—only put into a suitable form for the current exposition—by simply stating the definitions, while for a detailed explanation of the mathematical formalism the reader is referred to Allen et al. (2017) and Barrett et al. (2019). It is hard to overlook the large extent to which these definitions look formally analogous to the classical ones, however, it is important to stress that the analogy also has important limitations, which will be discussed later in Sect. 5.

In order to get some notational conventions out of the way, let $\mathcal{H}_S$ denote the Hilbert space (assumed to be finite-dimensional throughout for simplicity) associated with a

⁷ The case $P(Y, Z) = P(Y)P(Z)$ could be included seeing as $P(Y, Z)$ is Markov for the unconnected DAG with two vertices—we just tend to not find that case begging the question.

⁸ This condition is equivalent to the Markovianity of $P(Y, Z, X)$ for the ‘fork’ DAG in (c).

quantum system S and let $\rho_S \in \mathcal{L}(\mathcal{H}_S)$ denote the state of quantum system S , i.e. a density operator (positive semi-definite, trace-1). For the most general evolution of quantum systems, described by a completely positive (CP) trace-preserving map, for short CPTP map, $\mathcal{E} : \mathcal{L}(\mathcal{H}_A) \rightarrow \mathcal{L}(\mathcal{H}_B)$, we use a convenient representation through a variant of the Choi-Jamiołkowski (CJ) isomorphism (Choi, 1975; Jamiołkowski, 1972), whereby it is represented by a positive semi-definite and basis independent operator in $\mathcal{L}(\mathcal{H}_B \otimes \mathcal{H}_A^*)$, given by $\rho_{B|A} := \sum_{i,j} \mathcal{E}(|i\rangle_A \langle j|) \otimes |i\rangle_{A^*} \langle j|$, where $|i\rangle_A$ denotes an ortho-normal basis of $\mathcal{H}_A$ and $|i\rangle_{A^*}$ the corresponding dual basis of $\mathcal{H}_A^*$, the dual space of $\mathcal{H}_A$.

The *quantum common cause principle* can now be stated as:

QCCP Given two quantum systems B and C that are ‘correlated’, i.e. in a non-product state $\rho_{BC} \neq \rho_B \otimes \rho_C$,

- (0) there is a causal explanation of this fact, possibly involving a further system A , through either (a) $B \rightarrow C$, (b) $B \leftarrow C$, or (c) $B \leftarrow A \rightarrow C$, or else (c) in conjunction with (a) or (c) in conjunction with (b).
- (1) if system A is the complete common cause of systems B and C , i.e. no further direct common cause is missing and (a) and (b) are false, then the CPTP map $\rho_{BC|A}$ that describes their evolution, factorises as $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$.

Here, in writing $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$, the two factors $\rho_{B|A}$ and $\rho_{C|A}$ are the CJ operators of the maps that arise from marginalising over C and B , respectively, i.e. $\rho_{B|A} = \text{Tr}_C[\rho_{BC|A}]$ and $\rho_{C|A} = \text{Tr}_B[\rho_{BC|A}]$. Note that if $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$ holds then $\rho_{B|A}$ and $\rho_{C|A}$ necessarily commute (due to the hermiticity of the left-hand side (Allen et al., 2017)). Also note that writing $\rho_{B|A} \rho_{C|A}$ uses the typical convention of suppressing identity operators, i.e. it is short-hand for $(\rho_{B|A} \otimes \text{id}_C)(\rho_{C|A} \otimes \text{id}_B)$; we will use this convention throughout.

In answer to the three questions above, according to *QCCP*, first, it is quantum systems that play the role analogous to that of classical random variables in *RCCP*. Second, it is a CPTP map $\rho_{BC|A}$, describing the evolution from A to BC , that is constrained by the causal assertion analogously to the conditional probability distribution $P(Y, Z|X)$ in *RCCP*. Third, the constraint itself is the factorisation of $\rho_{BC|A}$ as $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$. Note that, as the notational reminiscence suggests, a conditional probability distribution $P(Y, Z|X)$ is (encodable as) a special case of a CPTP map $\rho_{BC|A}$, and the classical principle *RCCP* is, in a precise sense, a special case of the quantum analogue *QCCP*.

Now, what of the Bell scenario? Recall from Sect. 1 that from the perspective of quantum theory we know perfectly well what to point to as the common cause. The quantum common cause principle *QCCP* had better comply with this intuition. This is indeed the case, in a rather trivial sense. Let A_B and A_C be the two subsystems (e.g. electrons) of A that are prepared in a maximally entangled state $\rho_A = \rho_{A_B A_C}$ and then sent off to the space-like separated regions B and C . Let $\rho_{B|A_B}$ and $\rho_{C|A_C}$ be the respective CPTP maps that describe this transfer of the systems, where we use the same labels for the regions and the respective quantum systems. The state ρ_{BC} that gives rise to the Bell inequality violating correlations, denoted $P(O_B, O_C|M_B, M_C)$ in Sect. 2, arises from the evolution of the initial state ρ_A through a map $\rho_{BC|A}$. This map $\rho_{BC|A}$

is clearly of the form $\rho_{BC|A} = \rho_{B|A_B} \otimes \rho_{C|A_C} = (\rho_{B|A_B} \otimes \text{id}_{A_C})(\rho_{C|A_C} \otimes \text{id}_{A_B}) = \rho_{B|A} \rho_{C|A}$ —it does indeed satisfy the condition in *QCCP*.

Note that the condition in *QCCP* is however more general than a situation like $\rho_{BC|A} = \rho_{B|A_B} \otimes \rho_{C|A_C}$. It allows cases where $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$ holds, but where A may not factorise in a way that makes the commutation relation $[\rho_{B|A}, \rho_{C|A}] = 0$ trivial, i.e. $\rho_{B|A}$ and $\rho_{C|A}$ may both act non-trivially on A , rather than just on a tensor factor of A , and yet commute. An instructive example of that kind was given in Allen et al. (2017): the incoherent copy map $\rho_{BC|A}^{\text{copy}}$ with A , B and C all two-dimensional quantum systems, and defined by measuring A in basis $|0\rangle, |1\rangle$ and conditional on the outcome preparing the state $|0\rangle|0\rangle$ or $|1\rangle|1\rangle$, respectively, as the output state at BC . This may in fact be seen as a canonical example where one would like to say A is a complete common cause. Indeed, it is the case that $\rho_{BC|A}^{\text{copy}} = \rho_{B|A} \rho_{C|A}$ (Allen et al., 2017), although $\mathcal{H}_A$ is a two-dimensional space that cannot factorise non-trivially at all. Importantly, *QCCP* does demand a still very restrictive condition—most CPTP maps $\rho_{BC|A}$ do not factorise, i.e. generally $\rho_{BC|A} \neq \rho_{B|A} \rho_{C|A}$.

Thus far, both the choice of object constrained by the principle, as well as the mathematical constraint it involves may seem ad-hoc. However, before turning to justifying the principle, let us first complete the ride through definitions and cover the general framework.

3.2.2 The general quantum causal principle and quantum causal models

The way the formalism of quantum mechanics is typically used is one that manifestly depends on assumptions on spacio-temporal relations. For instance, two space-like separated systems A and B are described by a state ρ_{AB} in the tensor product space $\mathcal{L}(\mathcal{H}_A) \otimes \mathcal{L}(\mathcal{H}_B)$, while the evolution of a system C to a (possibly different) system D at some later time is described by a CPTP map $\mathcal{E} : \mathcal{L}(\mathcal{H}_C) \rightarrow \mathcal{L}(\mathcal{H}_D)$, or $\rho_{D|C}$ in CJ representation. This, however, is an obstacle if the aim is to spell out the constraints from causal structure, understood as the set of causal relations that hold amongst a set of systems, on their quantum description, but *independently* from the compatibility of those causal relations with any particular embedding of the quantum systems into some spacetime. In textbook quantum theory there just is no object that, akin to a probability distribution $P(X_1, \dots, X_n)$, always is the same sort of object no matter what the causal or spacio-temporal relations are. It has been emphasised in many works, in particular in Leifer and Spekkens (2013), that such a neutral formalism seems to be a prerequisite for the development of a quantum generalisation of causal models.

Various research programmes have over the past few decades produced similar formalisms for quantum theory that, although not all developed for the same purpose, can all be seen as meeting such a ‘neutrality’ requirement. They typically build on the CJ isomorphism employed above—maybe unsurprisingly so, seeing as the latter already has a CP map represented as a positive operator, similarly to states. The approach of Barrett et al. (2019) uses the *process formalism*,⁹ where the basic object is a

⁹ Up to slightly different conventions the same as process matrices, introduced in Oreshkov et al. (2012).

process operator $\sigma_{A_1 \dots A_n}$, for short process, over so called quantum nodes $A_1, \dots, A_n$. This is the second and last bit of conventions to get out of the way.

A *quantum node* A is the pair of a Hilbert space $\mathcal{H}_A$ and its dual $\mathcal{H}_A^*$ and a useful abstract concept that can come with different roles and interpretations. One is that a node corresponds to a ‘hypothetical cut’, a locus, where an agent’s intervention may or may not happen. On that reading the two Hilbert spaces are associated with the incoming and outgoing quantum systems before and after the intervention. Intuitively speaking, a *process operator* $\sigma_{A_1 \dots A_n}$ then is a generic means to represent the evolution of all considered systems outside the nodes.¹⁰ This includes in particular the above two paradigmatic situations of a multipartite state like ρ_{AB} and an ordinary CPTP map like $\rho_{D|C}$, possibly together with some initial state ρ_C , if in each case formally regarding A, B, C and D as defining quantum nodes. Technically speaking, $\sigma_{A_1 \dots A_n}$ is a positive operator on all involved Hilbert spaces $\bigotimes_{i=1}^n \mathcal{H}_{A_i} \otimes \mathcal{H}_{A_i}^*$ subject to further conditions that ensure it to be a well-defined and predictive empirical quantity.

For the details of the process formalism using these particular conventions see Barrett et al. (2021), Barrett et al. (2019) and references therein for related presentations. All that matters for the subsequent discussion are two facts. First, a process operator $\sigma_{A_1 \dots A_n}$ over quantum nodes $A_1, \dots, A_n$ can describe any situation on the same footing—regardless of the systems’ causal or spacio-temporal relations and regardless of whether they are open or closed. Second, given a process $\sigma_{A_1 \dots A_n}$ the joint probability distribution over the outcomes of any choice of interventions at the nodes $A_1, \dots, A_n$ can be predicted. The significance of this second fact for quantum causal reasoning will be discussed later on.

Now, taking inspiration from *QCCP*, the quantum analogue of the classical Markov condition that was first suggested and studied in the works of Allen et al. (2017) and Barrett et al. (2019), is as follows.

Definition 3.3 (Quantum Markov condition): Given a process $\sigma_{A_1 \dots A_n}$ and a DAG G with vertices $A_1, \dots, A_n$, the process is said to be Markov for G iff $\sigma_{A_1 \dots A_n}$ admits a factorization into pairwise commuting CJ operators (of corresponding CPTP maps) of the form $\sigma_{A_1 \dots A_n} = \prod_{i=1}^n \rho_{A_i|Pa(A_i)}$.

Importantly, the condition that the factors commute pairwise as stipulated in Def. 3.3 is a non-trivial and substantial part of the definition given that the parental sets generally overlap.¹¹ The corresponding *quantum causal principle* then takes the expected form:

QCP Given quantum nodes $A_1, \dots, A_n$ and a process operator $\sigma_{A_1 \dots A_n}$,
 (0) there is a causal structure of the quantum nodes $A_1, \dots, A_n$, possibly involving other (latent) quantum nodes as further common causes.

¹⁰ One may find the perspective of higher-order maps helpful: just as CP maps map states to states, processes map CP maps into the probabilities, where the input CP maps correspond to intervention outcomes or, in less operationalist terms, just some (non-deterministic) events at the nodes, including as special case ‘nothing happening at all’.

¹¹ While a simple consequence from the hermiticity of the factorising operator in the case of *QCCP*, commutativity of the factors does not follow from the hermiticity of $\sigma_{A_1 \dots A_n}$ when there are more than two factors.

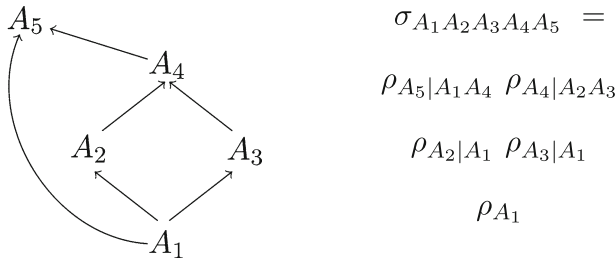


Fig. 3 A generic example of a quantum causal model

- (1) if the set $\{A_1, \dots, A_n\}$ is causally sufficient and the causal structure given by the DAG G with vertices $A_1, \dots, A_n$, then the process operator $\sigma_{A_1 \dots A_n}$ is Markov for G .

Without giving the details, note that *QCCP* is a special case of this general principle.¹² Now completing the analogy to the classical case, the definition of a quantum causal model from Allen et al. (2017) and Barrett et al. (2019) that is in keeping with *QCP*, is the following.

Definition 3.4 (Quantum causal model): A Quantum causal model (QCM) is given by:

- (1) a causal structure represented by a DAG G with vertices corresponding to quantum nodes $A_1, \dots, A_n$,
- (2) a process operator $\sigma_{A_1 \dots A_n}$ that is Markov for G .

Having laid out the main definitions and concepts of the framework of quantum causal models, at least at a formal level, it is the subsequent Sects. 4 and 5 that will discuss them, provide some context and try to build some intuition for these definitions.

4 Justification of the definitions

4.1 Some of the main results of the quantum framework

Let us begin by asking how this framework of quantum causal models from Allen et al. (2017) and Barrett et al. (2019), the basic definitions of which above Sect. 3.2 sketched, fare at a technical level and what results it has facilitated at all.

First, it should be noted that the classical framework is formally contained in the quantum one as a special case. Basically, a quantum causal model, the process operator of which is diagonal in a product basis of all Hilbert spaces, induces a classical causal model and vice versa. The precise way in which this holds is best understood via an intermediate kind of causal models—*classical split-node causal models* (Barrett et al., 2019)—which are insightful in their own right by bringing out more clearly which differences between the classical and quantum frameworks are down to differences

¹² A map $\rho_{BC|A}$ together with an initial state ρ_A defines a process operator over three nodes σ_{ABC} .

between classical probability theory and quantum theory proper and which are down to ‘matters of bookkeeping’ associated with the sort of data that appears in typical problems in either realm. See Barrett et al. (2019) for details.

Second, the framework facilitated a quantum generalisation of the *d-separation theorem*, which is one of the core results of the classical framework, capturing the link between causal structure and conditional independence relations in a probability distribution. As such, the generalised theorem (Barrett et al., 2019) relied on first introducing a generalisation of the notion of conditional independence in appropriate, genuinely quantum terms. The conception of this notion of so called *quantum relative independence* (Barrett et al., 2019) is novel and appears interesting in its own right. It was heavily inspired by having first identified the quantum analogue of Reichenbach’s principle, while at the same time and in contrast to the classical case, it cannot just be read off from it, given that for the general case it has to be a condition on a process operator.

Third, one of the main points of any causal model framework arguably would seem to be that it enables a principled *causal discovery* machinery. An immediate question here is what even constitutes the problem of quantum causal inference seeing as the explanandum that the respective classical and quantum causal model definitions address, changes substantially going from the former to the latter. But there are nonetheless interesting causal inference questions (see Sect. 5 for a more detailed comment; also see Bai et al., 2020; Barrett et al., 2019; Costa & Shrapnel, 2016; Giarmatzi & Costa, 2018). As mentioned in Sect. 3.1 the role of the classical causal principle *CP* in this context is its part (1) in the contrapositive form. Analogously, one would expect the same of its quantum analogue *QCP* and there indeed is a first, basic version of a quantum causal discovery algorithm sketched in Barrett et al. (2019) that relies on *QCP* in this expected way.

4.2 The right way round

From a conceptual point of view, one may, or even should, not be moved much by the above mentioned results. The most pertinent question must concern the justification of the causal principle—*why this quantum Markov condition?* Merely arguing from the above ‘successes’ of the quantum framework seems in danger of circularity. One would want to first establish confidence in the definition and *then* base e.g. a quantum causal discovery algorithm on it.

The same question can of course be raised in the classical case and the justification and status of Reichenbach’s common cause principle and of the causal Markov condition are the subject of ongoing discussions (see Hitchcock, 2020; Hitchcock & Rédei, 2020 for good overviews). However, there do exist various approaches to answering the question in the classical case, of which the following two are the most common.

One is to assume a deterministic macroscopic world with causal relations defined by functional dependencies, where probabilities enter the stage due to a lack of access to, or interest in the details of, certain variables. As is well known, one can then ground the causal Markov condition in *structural equational models* (SEMs) in the following sense. Suppose an SEM is given for variables $X_1, \dots, X_n$, i.e. a set of equations $\{X_i =$

$f_i(Pa(X_i), \lambda_i)\}_{i=1}^n$ where the function f_i fixes variable X_i in terms of its direct causes $Pa(X_i) \subset \{X_1, \dots, X_n\}$ and a ‘local disturbance’ (or ‘error variable’) λ_i that is not a direct cause of any of the other X_j ($j \neq i$). Assuming that the probability distribution $P(\lambda_1, \dots, \lambda_n)$ that encodes our lack of knowledge of these local disturbances, is a product distribution, i.e. the λ_i are all probabilistically independent from each other, then marginalisation over these local disturbances defines a probability distribution $P(X_1, \dots, X_n)$ that is Markov for the DAG defined by the parental sets $Pa(X_i)$. Conversely, given a $P(X_1, \dots, X_n)$ that is Markov for some DAG G , one can find an appropriate SEM that gives back $P(X_1, \dots, X_n)$ in the way as above and that has the sets of direct causes $Pa(X_i)$ in the functions $f_i : Pa(X_i) \times \lambda_i \rightarrow X_i$ coincide with the respective parental sets in the DAG G (Pearl, 2009). In this approach, which is the one Pearl takes (Pearl, 2009; Pearl & Verma, 1995) (also see (Hitchcock, 2020)), the causal Markov condition, or more precisely part (1) of *CP*, is a theorem rather a mere stipulation.

Another approach is to see the causal Markov condition as having a more normative role, supported by the fact that there are many cases where we find it uncontroversial to assert what the ‘correct causal explanation’ is and these examples are indeed in keeping with the causal Markov condition according to *CP* (Hitchcock, 2020; Spirtes et al., 2000). Given its empirical success, the question after a justification may not necessarily turn on foundational questions on the sort of definition of the causal relation that may be seen to underlie it and what the metaphysical status of causal relations is (Beebe et al., 2009; Eberhardt, 2009).

In the quantum case, however, there does not exist an analogous set of uncontroversial examples, in which everyone agreed on the appropriate respective causal explanations. An analogue to this second approach thus does not seem available and the question after a justification or conceptual clarification hence looms large.

4.3 Definition of the quantum causal relation

In fact at the heart of the works of Allen et al. (2017) and Barrett et al. (2019) is a proposal to address the question after a justification of the definitions of the quantum framework and in particular of *QCP*. The main idea is to give a mathematical definition of the quantum causal relation, which then opens the possibility of obtaining the principle as a theorem, rather than leaving it an ad-hoc claim, similarly to the SEM approach to classical causal models mentioned above. Here, providing a definition of quantum causal relations is not to the end of moving away from causal relations as primitive notions by analysing them in terms of relations of counterfactual dependencies or some other type of relation, but to the end of formalising causal intuitions, in a way that is consistent and well-motivated from the structure of quantum theory as a theory of fundamental physics.

The common cause case

This proposal defines causal relations in terms of influence through unitary evolution, where influence from C to E through a unitary map U means that the marginal output state ρ_E depends on the input state ρ_C , at least for some assignment of states to all other input systems. More formally,

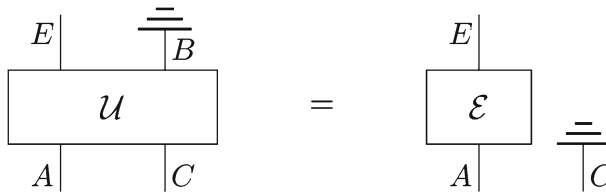


Fig. 4 Diagrammatic representation of the condition that $C \nrightarrow E$ for a given unitary map $U : \mathcal{H}_A \otimes \mathcal{H}_C \rightarrow \mathcal{H}_E \otimes \mathcal{H}_B$, i.e. a representation of the equation $\text{Tr}_B[U(_)U^\dagger] = \mathcal{E} \otimes \text{Tr}_C(_)$ for some CPTP map $\mathcal{E}$. Here the ‘upside-down grounding symbol’ represents the partial trace and $U(_) := U(_)U^\dagger$ denotes the CPTP map that arises from U . (See, e.g., Coecke & Kissinger 2017 for an introduction to this diagrammatic language for quantum processes.) Note that the condition $C \nrightarrow E$ is furthermore equivalent to that $\text{Tr}_B[\rho_{EB|AC}^U] = \rho_{E|A}^\mathcal{E} \otimes \mathbb{1}_{C^*}$ when written in terms of the CJ operators $\rho_{EB|AC}^U$ and $\rho_{E|A}^\mathcal{E}$ of U and $\mathcal{E}$, respectively

Definition 4.1 (No-influence and direct cause relation): Given a unitary transformation $U : \mathcal{H}_A \otimes \mathcal{H}_C \rightarrow \mathcal{H}_E \otimes \mathcal{H}_B$, write $C \nrightarrow E$, say ‘ C does not influence E ’, if and only if there exists a CPTP map $\mathcal{E} : \mathcal{L}(\mathcal{H}_A) \rightarrow \mathcal{L}(\mathcal{H}_E)$, such that $\text{Tr}_B[U(_)U^\dagger] = \mathcal{E} \otimes \text{Tr}_C(_)$. System C is said to be a direct cause of E , written $C \rightarrow E$, if and only if C can influence E , i.e. $\neg(C \nrightarrow E)$.

If causal relations pertain to only unitary transformations, which thereby *have* a causal structure by definition, how is it that they constrain a generally non-unitary CPTP map according to *QCCP*? The idea here is that evolution is fundamentally unitary and a CPTP map $\rho_{BC|A}$ is said to be *compatible* with A being a complete common cause of B and C iff it can be seen to arise from an (underlying) unitary map, where A indeed is the complete common cause as far as B and C are concerned. More formally, $\rho_{BC|A}$ is compatible with A being a complete common cause of B and C iff there exists a unitary map $U : \mathcal{H}_{\lambda_B} \otimes \mathcal{H}_A \otimes \mathcal{H}_{\lambda_C} \rightarrow \mathcal{H}_B \otimes \mathcal{H}_F \otimes \mathcal{H}_C$, generally involving some additional systems λ_B , λ_C and F (systems in the ‘environment’), such that λ_B and λ_C are not further common causes of B and C , i.e. $\lambda_B \nrightarrow C$ and $\lambda_C \nrightarrow B$ through U , and such that the map $\rho_{BC|A}$ is obtained from U when fixing some product state $\rho_{\lambda_B} \otimes \rho_{\lambda_C}$ for the auxiliary input systems and ignoring the additional output system F .

Importantly, any CPTP map, in particular one of the form as $\rho_{BC|A}$, always has a (non-unique) ‘unitary dilation’, which means that it can be seen to arise from some unitary map by ignoring some additional output system and fixing an initial state for an additional input system. However, this generally requires the involvement of additional systems that do play the role of further common causes. Compatibility with the complete common cause scenario hence is a non-trivial condition. Furthermore, note that part of this notion of compatibility is that the additional local disturbances λ_B and λ_C are uncorrelated for they are assumed to be in a product state. This assumption is completely analogous to the common assumption of uncorrelated local disturbances in classical SEMs, mentioned in Sect. 4.2. It is nonetheless a substantial assumption and while the intuition behind it is clear—if it were necessary to consider additional λ_B and λ_C in a *non-product* state, it would seem this must be because ‘further down the line’ there is a common cause in turn responsible for that fact, and this common

cause should hence be included—it is not straightforward how to justify this product state assumption in a way that does not end up being circular.

The main theorem from Allen et al. (2017) establishes equivalence between $\rho_{BC|A}$, on the one hand, being compatible with A being a complete common cause of B and C and, on the other hand, factorising as $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$. Hence, whenever according to Def. 4.1 one would say A is a complete common cause of B and C then the consequent of principle *QCCP* indeed necessarily holds and conversely, whenever that condition holds one can stipulate this to be because of the alleged causal relations. The condition in *QCCP* could hence not be stronger.

The general case

Barrett et al. (2019) generalises this approach to arbitrary DAGs. To this end, the definition of direct-cause relations between quantum systems relative to a unitary transformation in Def. 4.1 is lifted to one between quantum nodes relative to a ‘unitary process’. We refer the reader to Barrett et al. (2019) for the details, but, roughly speaking, a unitary process in this context can be seen as the sort of object obtained in the following way. Suppose a unitary map U is given that describes the evolution of some set of systems (possibly into some set of other systems); this map U may have a complex compositional structure reflecting the interaction between subsets of the systems, the transformation into various intermediate systems, and possibly several time steps—that is, in general, U may be given as the composition, sequentially and in tensor product, of various component unitary maps¹³; then formally introduce ‘cuts’ in arbitrary places, possibly involving some of the intermediate systems, to define quantum nodes as the relata for the causal analysis; the resultant object is what is called a unitary process. The notion of causal influence at the process level can then be defined in exactly the same spirit as in Def. 4.1 and unitary processes then by definition *have* a causal structure.

Generalising the simple common cause case above, it is unitary processes that are now assumed to underlie any kind of evolution described by non-unitary processes. A process $\sigma_{A_1 \dots A_n}$ is said to be ‘compatible with a DAG G ’ if there exists a unitary process from which $\sigma_{A_1 \dots A_n}$ can be seen to arise and, crucially, such that the unitary process does not introduce additional common causes to the nodes $A_1, \dots, A_n$ that are not already present in G . Barrett et al. (2019) establishes equivalence between compatibility with and Markovianity for a DAG, thereby putting the principle *QCP* and the notion of a quantum causal model from Def. 3.4 on a clear conceptual ground.

As already mentioned earlier, this approach is analogous to the SEM approach to classical causal models that obtains the classical causal Markov condition as a theorem. However, there it is on the basis of causal relations defined as functional dependencies. While the assumption of a macroscopic determinism may or may not appeal to one, the special role of unitary dynamics within quantum theory for describing the evolution of a closed set of systems is little controversial.

What of signalling?

¹³ In short, U may be given as the composition of other unitary maps in a way that is representable as a circuit diagram (Coecke & Kissinger, 2017).

Section 2 mentioned a plethora of works on causality in quantum foundations and quantum information theory with the predominant intuition being ‘causation as signalling’. How then does the above sketched proposal from Allen et al. (2017) and Barrett et al. (2019) relate to that and provide the clarification that the introduction in Sect. 1 called for?

First, given a unitary map, if input system C is a direct cause of output system E according to Def. 4.1, it is in particular true that an agent with access to C can in principle send signals to an agent with access to E . Similarly for the more general picture of direct-cause relations between quantum nodes relative to unitary processes (Barrett et al., 2019). Importantly, there are however cases where the converse is not true in the sense that the impossibility to signal does not imply the absence of a causal relation, at least not without further qualification. See Barrett et al. (2019, 2021) for concrete examples, but the argument is that there are cases where it is uncontroversial—that is, in keeping with the very intuition of ‘causation as signalling’—to claim that A is a direct cause of B given a particular unitary evolution of a number of systems, but if some, possibly inaccessible systems in the environment happen to be in a particular (‘fine-tuned’) state, it is factually impossible for any agent to send signals from A to B .

This demonstrates that the possibility to signal is *not* a satisfactory definition unless the kind of data relative to which signalling is considered is further qualified to be a particular one—which, however, it usually is not in the literature. The possibility of signalling from A to B , as such in an unqualified way, is thus a sufficient but not necessary condition for the presence of a causal relation. This is of course not surprising at all, seeing as it has its well-known, purely classical analogue in the causal model literature, surfacing in particular in the discussion of fine-tuning and the notion of faithfulness from an SEM perspective.

The schematic structure of the approach

One may summarise the approach as:

- (1) Define the causal relation as a 3-place relation of the form ‘ C is a direct cause of E given $\mathcal{M}$ if and only if some condition CR holds’, that is, it has to say what sort of object the relata C and E are, relative to what sort of data $\mathcal{M}$ causal relations are defined at all and what the causal connection is, captured in the condition CR .
- (2) Spell out how to give causal explanations, in accordance with (1) and in a principled way, when the given data is not of the kind as demanded by $\mathcal{M}$.

The SEM-based approach to classical causal models answers (1) with classical variables as relata, a set of functions as $\mathcal{M}$ and functional dependency as the causal connection. The approach to quantum causal models from Allen et al. (2017) and Barrett et al. (2019, 2021) answers (1) with quantum systems (more generally quantum nodes) as relata, a unitary map (unitary process) as $\mathcal{M}$ and relations of influence between input and output systems as the causal connection.

What then constitutes a causal model framework in particular is an answer to (2), i.e. the framework has to address the epistemological problems arising from the fact that the data one typically encounters is not of the kind $\mathcal{M}$ that appears in (1). The precise way in which the quantum causal model framework addresses (2) is inscribed into the

causal principle *QCP*, in particular the role of the quantum causal Markov condition. This condition is indeed in accordance with (1) for it is ‘obtained as a theorem’ on the basis of the answers to (1).

5 Quantum vs classical causal explanations

The relation between classical and quantum causal models, including that between classical and quantum causal principles, can be told as a story of similarities as much as one of differences. The presentation of the basic definitions in Sect. 3 focused on the analogous forms the definitions take and how the quantum versions subsume the classical ones. Furthermore, Sect. 4 observed the analogy between the sketched approach of justifying the quantum definitions and an SEM approach to classical causal models. As for the differences, it is worth reiterating and emphasising a couple of points in the following.

Probability distribution vs process operator

The first concerns the respective sorts of objects constrained by causal structure. Classically, a probability distribution $P(X, Y, \dots)$, suppose estimated from observational data, allows to evaluate Bayesian conditionals $P(Y|X)$ to predict the probability for observing $Y = y$ in case $X = x$ is observed. Just $P(X, Y, \dots)$ on its own, however, does not allow one to say anything about Y if one intervened on X and, say, forced it to satisfy $X = x$. In contrast, if supplied with causal rather than merely statistical knowledge one may be able to evaluate claims about such interventional scenarios. Indeed, if given a causal model for $P(X, Y, \dots)$, then for any intervention on a subset of the variables that are included in the model the consequences of such intervention for the remaining variables can be handled. This is a paradigmatic usecase of the classical framework; see, e.g., Bareinboim et al. (2020) for a formal analysis of what kinds of queries can be answered on the basis of what kinds of data.

Similarly, given a quantum causal model involving process operator $\sigma_{A_1 \dots A_n}$, one can make predictions for the outcomes of interventions at any of the quantum nodes $A_1, \dots, A_n$, just as one would expect of a causal model framework. However, as noted in Sect. 3.2, a process operator $\sigma_{A_1 \dots A_n}$ alone already has that ‘full predictiveness’ for interventions, even if it is not part of a quantum causal model.¹⁴ A process operator $\sigma_{A_1 \dots A_n}$ is an (in principle) empirically accessible object, but can only be estimated from interventional data. In fact, it requires so called ‘informationally complete interventions’ at all nodes $A_1, \dots, A_n$, which makes it practically unfeasible unless n and the dimensions of all involved systems are small. Should one be surprised that $\sigma_{A_1 \dots A_n}$ plays the analogous role to that of a probability distribution for a classical causal model? It seems not, seeing as in quantum theory there just is no notion of an ‘observation’. Any measurement that is informative is necessarily disturbing, i.e. it is an intervention. As far as an object goes that is formulated in intrinsically quantum

¹⁴ Note that a process $\sigma_{A_1 \dots A_n}$ may not be Markov for *any* DAG with vertices $A_1, \dots, A_n$ and therefore not even admit a quantum causal model if the set of quantum nodes is restricted to $A_1, \dots, A_n$ (Barrett et al., 2019).

terms, that is empirical and that is ‘neutral’ in the sense as discussed in Sect. 3.2, a process operator is precisely that.

An immediate question is what a problem of quantum causal discovery would then consist in if not defined by given *observational* data. Note that even given a process operator $\sigma_{A_1 \dots A_n}$ with which one can check from which subsets of nodes signalling is possible to which other subsets of nodes, there are non-trivial questions: what is the condition for when that ‘signalling pattern’ can be taken to be a plausible causal explanation? When are common causes, not included in $A_1, \dots, A_n$, necessarily missing from the picture?¹⁵ All of these seem pretty proper causal inference questions—ones that can only be answered meaningfully and in a principled way on the basis of a quantum causal model framework.¹⁶ As mentioned earlier, Barrett et al. (2019) sketches a very basic quantum causal discovery algorithm that allows to answer these questions in some cases. It also argues for why the quantum causal model framework cannot but be the foundation for also more practically relevant versions of causal inference problems defined by partial interventional data, from which $\sigma_{A_1 \dots A_n}$ cannot be inferred uniquely (which is distinct from knowing a process operator perfectly but just on a subset of a causally sufficient set of nodes). It is future developments, for instance in quantum computing and more generally, quantum information processing, that will have to inform what really are the practically relevant problems here.

Same but different—factorisations marking successful causal explanations

The second point on classical-quantum differences concerns the explanans, more concretely the condition that witnesses the plausibility of a candidate causal explanation and on what our confidence in that condition rests. To this end, let us revisit the complete common cause case. Classically, the explanandum is the correlation between variables, say, Y and Z . Once identified a candidate complete common cause X and provided none of Y and Z are causes of each other, the criterion according to *RCCP* that Y and Z ought to be independent conditional on X to mark a successful common cause explanation, really is rather intuitive—the thing that supposedly is responsible for underwriting correlations in the first place is held fixed, hence, there then should be no further association between Y and Z . This is to say, regardless of how much credence one has in any attempt of defending it against various challenges (Hitchcock & Rédei, 2020) and in any attempt of justifying Reichenbach’s common cause principle, such as from an underlying determinism, the essence of the criterion is intuitive in its own right.

The situation is different in the quantum case. The analogue of what begs the question is a non-product state $\rho_{BC} \neq \rho_B \otimes \rho_C$, a fact that is equivalent to that at least for some choices of measurements at B and C the outcomes will be correlated (though this of course may or may not violate a Bell inequality). Suppose $\rho_{BC|A}$ is the CPTP map (in CJ representation) that gives rise to the state ρ_{BC} if letting system A evolve

¹⁵ Given the process formalism’s generality, there also is the question of whether a given $\sigma_{A_1 \dots A_n}$ is compatible with the existence of a causal order of its nodes at all (Barrett et al., 2021; Costa & Shrapnel, 2016).

¹⁶ Similarly and in a purely classical context, causal inference is not just turned into a trivial matter if one has access to interventions on (some of) the variables and there has been extensive work on systematising causal inference in such situations (see, e.g., Eberhardt & Scheines 2007; Hyttinen et al. 2013).

according to $\rho_{BC|A}$ from initial state ρ_A . The criterion for a successful common cause explanation according to *QCCP* demands that $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$ holds. First, it is $\rho_{BC|A}$ that factorises, while the outcome probabilities for measurements at B and C will generally remain correlated even when conditional on the fact that A is a complete common cause and that it was in state ρ_A . This just reflects the turn that the presented research programme took at the very beginning as explained in Sect. 1, namely that it did not aspire to find a Reichenbachian common cause that would show itself in a factorising (even if generalised) probability distribution. Second, finding the condition $\rho_{BC|A} = \rho_{B|A} \rho_{C|A}$ as such intuitive is not as straightforward as in the classical case. Rather, it is the definition of quantum causal relations from which the credence in that condition stems, recalling the condition's equivalence to the *compatibility* of $\rho_{BC|A}$ with A being the complete common cause of B and C .

6 Note on causal vs temporal asymmetry

Reichenbach's common cause principle appeared as part of the (posthumously published) book *The Direction of Time* (Reichenbach, 1956), which is concerned with the relation between various temporal asymmetries. It might thus be expected that a work on a quantum common cause principle claiming to take inspiration from Reichenbach's principle, should comment on its relation to the arrow(s) of time. Also, in the spirit of the Russellian metaphor of the lurking danger of the monarchy (Russell, 1912), a reader may wonder how the perspective from quantum theory as a fundamental, time-reversal invariant theory in physics can go together at all with any serious scientific interest in causation. The present paper, however, does not pretend to have anything substantial to say on such matters as the status of the arrow(s) of time in physics, let alone its relation to an alleged causal arrow. It contents itself with the following remark to hint at a natural direction.

The proposal sketched in Sect. 4 has quantum causal relations inhere in unitary transformations. Any unitary map $U : \mathcal{H}_{C_1} \otimes \dots \otimes \mathcal{H}_{C_n} \rightarrow \mathcal{H}_{D_1} \otimes \dots \otimes \mathcal{H}_{D_k}$ from some n systems to some k (possibly different) systems has a causal structure given by the set of direct-cause relations of the form $C_i \rightarrow D_j$ and is representable by a DAG G (Barrett et al., 2019).¹⁷ Now, the inverse map $U^\dagger$ is also unitary and associated with time-reversed evolution. Crucially, its causal structure is precisely that obtained by inverting all arrows of G (Arrighi et al., 2011; Lorenz & Barrett, 2021),¹⁸ i.e. $C_i \rightarrow D_j$ relative to U implies $D_j \rightarrow C_i$ relative to $U^\dagger$. In this sense then one may say causal structure is symmetric, i.e. the 'causal connections' of which systems

¹⁷ Note that it is not an obvious and trivial observation that a unitary map has a causal structure representable as a DAG—this is a fact shown in Barrett et al. (2019). For a generic CPTP map, if some input system C cannot signal to some output system D and also not to some other output system D' , then it does not follow that C cannot signal to the composite system DD' and, hence, the 'signalling structure' of a CPTP map is in general not representable by a DAG. In contrast, given a unitary map, the 'single system' influence relations fully determine the overall structure of causal influence and the causal structure therefore is representable by a DAG.

¹⁸ Note that this is not entirely obvious as, for instance, an analogous property does not hold for classical reversible functions if defining causal relations in terms of functional dependencies (Lorenz & Barrett, 2021).

can influence which other systems can be considered as undirected, seeing as it is always the same pairs of systems standing in that causal relation, no matter which ‘way round’ one is looking at the unitary map. The research programme may thus lend support to a view that avoids debunking causation and has the essence of causal relations as symmetric and fundamental, while the directedness in identifying C_i to be the cause of D_j , rather than the other way around, comes down to the same origin as what brings about the temporal asymmetry.

This direction of thought in fact aligns well with recent efforts by a number of researchers on pinning down where and how the apparent temporal asymmetry enters the stage in the way the quantum formalism is used, given the time-reversal invariance present at the fundamental level of the theory (Chiribella & Liu, 2022; Di Biagio et al., 2021).

7 Conclusion

This paper discussed the framework of quantum causal models as developed in the recent works of Allen et al. (2017) and Barrett et al. (2019). While it did not engage with several further technical results that were presented in Barrett et al. (2019), it emphasised, analysed and argued for two main points. First, that the developments are a continuation of the spirit of, and a generalisation of, Reichenbach’s common cause principle when read as a link that guides causal reasoning by bridging between causal claims and the empirical description of the matter of affairs. Second, that the programme’s core of obtaining the quantum causal Markov condition as a theorem on the basis of a definition of the quantum direct-cause relation makes the work interesting for both quantum foundations and philosophy of causation. This framework then, while humble insofar as it has only thus far considered finite-dimensional systems, is also ambitious in that it strives to find out whether causal relations can consistently be taken as fundamental at a quantum level, and in what sense causal reasoning can be upheld when it comes to quantum systems.

The existence of Bell inequality violating quantum correlations together with the causal nature of the assumptions that go into Bell’s theorem are well-known reasons for contemplating the need and the conceivability of quantum causal reasoning. However, it cannot be overemphasized that the discussed approach of quantum causal models does not do away with the fact that quantum physics challenges classical intuitions—if anything it embraces that fact. It also does not provide a totally novel causal explanation of Bell correlations. Instead, it provides the grounds on which the intuition concerning what the obvious quantum common cause in the Bell scenario should be—coming from a quantum theoretical description of the situation—now becomes part of a rigorous and principled general causal reasoning framework. What the framework maintains is the possibility to give a causal account of such and, more generally, any quantum mechanical situation. What it gives up is that the factorisation condition that marks the successful causal explanation is a conditional independence condition in the terms of a (even if generalised) probability space, and instead has a CPTP map, or more generally, a process operator as the terms for the factorisation condition.

Beyond bringing value due to its clarificatory role to the active research area on quantum causal structure in quantum foundations and quantum information theory, the approach has arguably already paid off by facilitating further, closely related developments. These include work on the relation between causal and compositional structure of unitary transformations in Lorenz and Barrett (2021). The latter studies the insights and the consequences of an algebraic nature from the commutation relations that are dictated by causal structure as can be seen, for instance, in the quantum Markov condition (cf. Def. 3.3). The developments also include an extension of the quantum causal model framework to directed graphs with directed cycles in Barrett et al. (2021) and Ormrod et al. (2022) as a means to analyse in a new light processes with an ‘indefinite causal order’ that were mentioned in Sect. 2.

The framework of classical causal models can be employed in a great range of different situations and disciplines with varying kinds of relations—from receiving medical treatment to collisions of billiard balls; readings of barometers to tax parameters in economics—and with different underlying views on what probabilities are. The framework of quantum causal models is of course much more specific—derived from a particular scientific theory—but to what extent is it general in terms of its compatibility with different interpretations of quantum theory? And can one develop the relation between a causal and a temporal asymmetry that Sect. 6 alluded to, in a more detailed and precise way? Such questions, among many others, deserve further investigation. As set out in the introduction, this paper hopes, above all, to be an invitation for cross-disciplinary interaction and further philosophical analysis.

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Declarations

Conflicts of interest The author declares that he has no conflict of interest.

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